

2.1 NEWTON'S METHOD

- REQUIRES "GOOD", INITIAL GUESS, $x^{(0)}$: "GOOD" DEPENDS ON SPECIFIC NONLINEAR EQN BEING SOLVED
- ASSUME WE HAVE GOOD $x^{(0)}$
- ONE GENERAL STRATEGY: $x^{(0)}$ IS SOLUTION OF RELATED PROBLEM, PERHAPS WITH A SLIGHTLY DIFFERENT PARAMETER E.G.

DERIVATION OF NEWTON'S METHOD (TAYLOR SERIES)

- NEWTON'S METHOD REQUIRES, $f'(x)$ OF $f(x)$
- LET x^* BE A ROOT OF $f(x) = 0$ THEN TAYLOR EXPANDING

$$0 = f(x^*) = f(x^{(n)}) + (x^* - x^{(n)}) f'(x^{(n)}) + O((x^* - x^{(n)})^2)$$

EXPANSION POINT IS $x^{(n)}$

$$0 \approx f(x^{(n)}) + (x^* - x^{(n)}) f'(x^{(n)})$$

1) TREAT THIS AS EQN

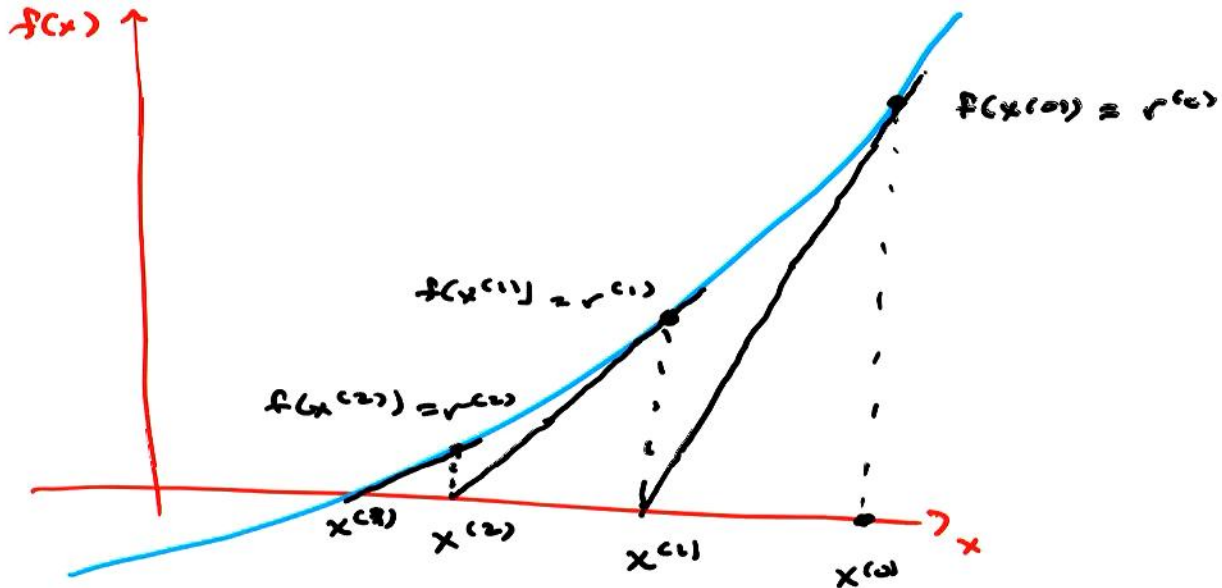
2) REPLACE x^* WITH NEW ITERATE $x^{(n+1)}$

$$0 = f(x^{(n)}) + (x^{(n+1)} - x^{(n)}) f'(x^{(n)})$$

OR

$$x^{(n+1)} = x^{(n)} - \frac{f(x^{(n)})}{f'(x^{(n)})} \quad (6)$$

NEWTON'S METHOD IN 1-D



- CAN WRITE ITERATION IN FORM

$$x^{(n+1)} = x^{(n)} - \frac{f(x^{(n)})}{f'(x^{(n)})} \quad (7)$$

$$f'(x^{(n)}) \delta x^{(n)} = f(x^{(n)}) = r^{(n)} \quad (8)$$

- NOTE THAT WE HAVE

$$f'(x^{(n)}) = \frac{f(x^{(n)})}{x^{(n+1)} - x^{(n)}} = \frac{\text{rise}}{\text{run}}$$

- CONVERGENCE: WHEN NEWTON'S METHOD CONVERGES IT DOES SO RAPIDLY: EXPECT NUMBER OF SIGNIFICANT DIGITS TO ROUGHLY DOUBLE AT EACH ITERATION (QUADRATIC CONVERGENCE)

- EXAMPLE: "SQUARE ROOTS"

$$f(x) = x^2 - a \rightarrow x^* = \sqrt{a} \quad (9)$$

APPLICATION OF (6) GIVES

$$x^{(n+1)} = x^{(n)} - \frac{x^{(n)2} - a}{2x^{(n)}}$$

$$= \frac{2x^{(n)2} - (x^{(n)2} - a)}{2x^{(n)}}$$

$$= \frac{x^{(n)2} + a}{2x^{(n)}}$$

CAN REWRITE AS

$$x^{(n+1)} = \frac{1}{2} \left(x^{(n)} + \frac{a}{x^{(n)}} \right)$$

• TRY IT "MANUALLY" COMPUTE

$\sqrt{2} = 1.414\ 2135\ 6237$ USING 12-DIGIT
ARITHMETIC (HAND CALCULATION)

ITERATE

XC FIGS

$$x^{(0)} = 1.5$$

$$x^{(1)} = \frac{1}{2} (1.5 + 2.0/1.5) = 1.416\ 6666\ 6667$$

$$x^{(2)} = \frac{1}{2} (1.416... + 2.0/1.416...) = 1.414\ 2156\ 8628$$

$$x^{(3)} = \frac{1}{2} (1.4142... + 2.0/1.4142...) = 1.414\ 2135\ 6237$$

QUADRATIC CONVERGENCE AS ADVERTISED

2. NEWTON'S METHOD FOR SYSTEMS OF EQUATIONS

• NOW WANT TO SOLVE

$$\underline{f}(\underline{x}) = \underline{0}$$

(1.1)

WHERE

$$\underline{x} = (x_1, x_2, \dots, x_d)$$

$$\underline{f} = (f_1(\underline{x}), f_2(\underline{x}), \dots, f_d(\underline{x}))^T$$

NOTE: $\underline{f}$ AND RESIDUAL VECTOR $\underline{r}$ ARE TO BE VIEWED AS COLUMN VECTORS (NOTATION MAY GET SLIPPERY)

• EXAMPLE ($d=2$)

$$\sin(x_1) = \frac{1}{2} \rightarrow \sin(x_1) - \frac{1}{2} = 0$$

$$y^2 = 6x + 2 \rightarrow y^2 - 6x - 2 = 0$$

• CANONICAL NOTATION

$$\underline{x} \equiv (x, y)$$

$$\underline{f} \equiv (f_1(\underline{x}), f_2(\underline{x}))$$

$$f_1(\underline{x}) = f_1(x, y) = \sin(x_1) - \frac{1}{2}$$

$$f_2(\underline{x}) = f_2(x, y) = y^2 - 6x - 2$$

• METHOD AGAIN ITERATIVE, START WITH $\underline{x}^{(0)}$ THEN GENERATE ITERATES

$$\underline{x}^{(0)} \rightarrow \underline{x}^{(1)} \rightarrow \dots \underline{x}^{(n)} \rightarrow \underline{x}^{(n+1)} \rightarrow \dots \underline{x}^*$$

(SOLN OF (1))

• DETERMINING $\underline{x}^{(0)}$ EVEN HARDER FOR 1-DIM. CASE (BUT USE CONTINUATION IF POSSIBLE) ASSUME THAT $\underline{x}^{(0)}$ IS "GOOD"

• ANY ESTIMATE $\underline{x}^{(n)}$ HAS AN ASSOCIATED RESIDUAL

$$\underline{r}^{(n)} \equiv \underline{f}(\underline{x}^{(n)}) \quad (2)$$

- ANALOGUE OF $f'(x)$ HERE IS THE JACOBIAN MATRIX
 $\underline{J}$ OF FIRST DERIVATIVES WITH ELEMENTS J_{ij}

$$J_{ij} = \frac{\partial f_i}{\partial x_j} \quad (3)$$